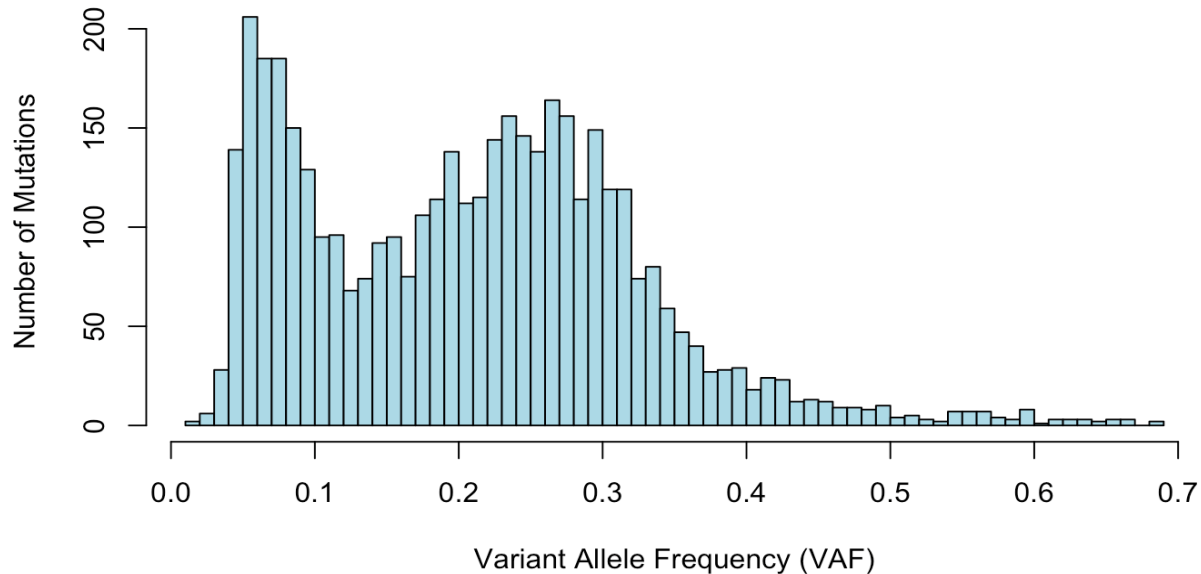


mClust report:
How to create the cancer phylogenetic tree ?

For the single sample analysis,

1. Load the “LU-4.snv.txt” file, and run the mClust single sample clustering analysis using variant allele frequency (VAF) of all mutations identified. First, generate a histogram of all VAFs, and briefly comment on the VAF distribution (**2 marks**).

Distribution of Variant Allele Frequencies in LU-4 Tumor Sample



Observations:

The VAF distribution is multimodal, with:

- a low-VAF peak between ~ 0.05 – 0.1
- a broader peak between ~ 0.20 – 0.30
- a right-hand tail with local density increases around ~ 0.35 – 0.4 , extending up to ~ 0.6

Interpretation/ comment:

The VAF histogram shows a bimodal distribution, indicating the presence of multiple mutation populations rather than a single dominant clonal peak. In contrast to a unimodal distribution, where tumour purity can be approximated directly from the mean or median VAF, the presence of multiple peaks reflects a mixture of clonal, subclonal, and potentially homozygous/LOH-associated mutations. Therefore, tumour purity cannot be reliably estimated from the raw VAF histogram alone and requires separation of mutation clusters using model-based clustering.

2. Run the density estimation analysis via Gaussian finite mixture modelling, followed by dimension reduction clustering and classification analysis. How many clusters can you identify? What is the size (# of mutations) for each cluster? What is the mean VAF value for each cluster? Can you identify which are major clones (heterozygous or homozygous), which are subclones, and their associated cancer cell fractions (CCF)? You can summarise all these into a table as shown in the lecture/practical slides (3 marks).

- To characterise tumour clonality in LU-4, variant allele frequencies (VAFs) were analysed using Gaussian finite mixture modelling with mClust, followed by dimension-reduction clustering and classification. This approach allows separation of mutations into distinct clusters with similar VAFs, enabling identification of major clonal and subclonal populations. Based on the dominant heterozygous clonal peak, tumour purity was inferred and used to estimate cancer cell fractions (CCF) for each cluster, allowing classification into clonal (heterozygous or homozygous/LOH-like) and subclonal groups.
- Based on the size, cluster 3 was identified as the dominant heterozygous clonal cluster. Four clusters were identified.**

Dimension reduction for model-based clustering and classification

Mixture model type: Mclust (V, 4)

```
Clusters      n
      1    619
      2    600
      3   2823
      4    165
```

Estimated basis vectors:

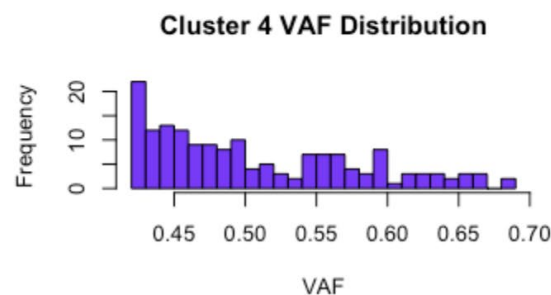
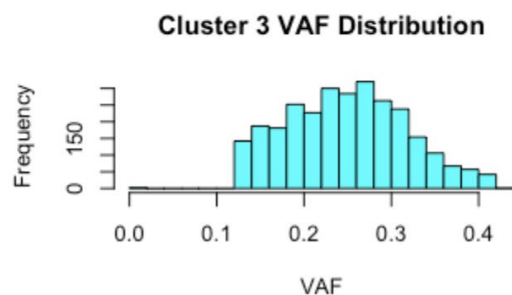
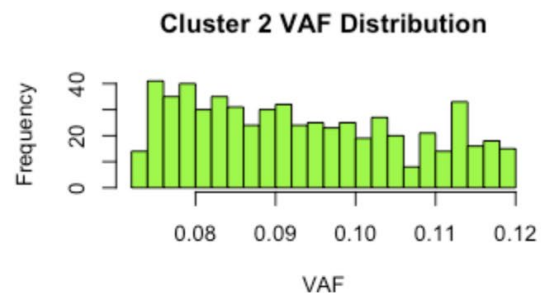
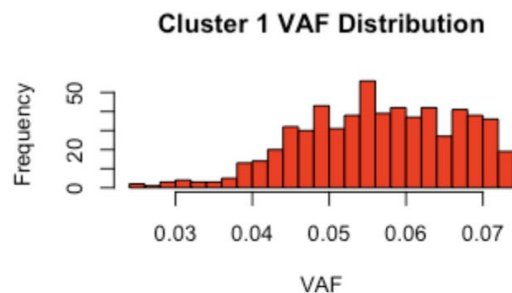
```
          Dir1
data1$vaf    1
```

```
          Dir1
Eigenvalues  0.88272
Cum. %      100.00000
```

Cluster <dbl>	Mean VAF <dbl>
1	0.05643398
2	0.09312967
3	0.24892659
4	0.50731950

CCF values were calculated by dividing each cluster's mean VAF by the mean VAF of the dominant heterozygous clonal cluster (Cluster 3), with values capped at 1.

Gaussian mixture modeling identified **four distinct VAF clusters**, indicating intratumour heterogeneity in the LU-4 sample.

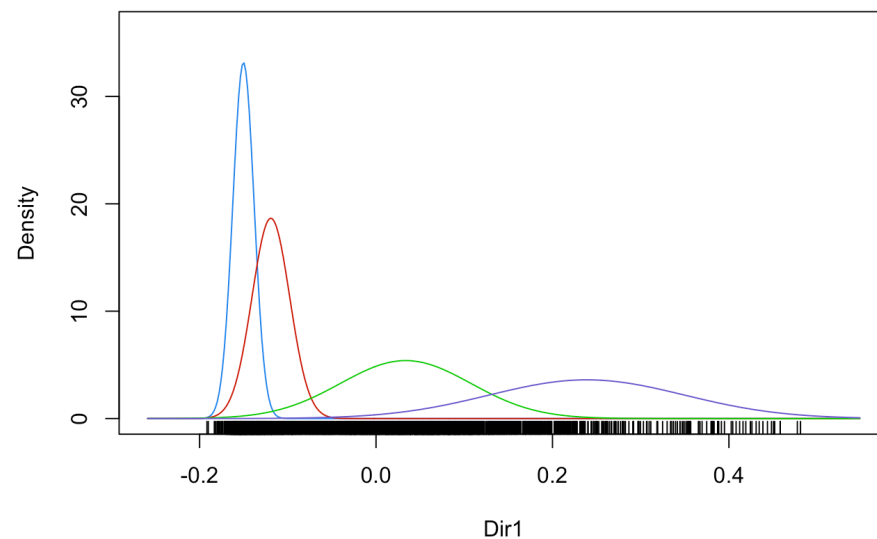
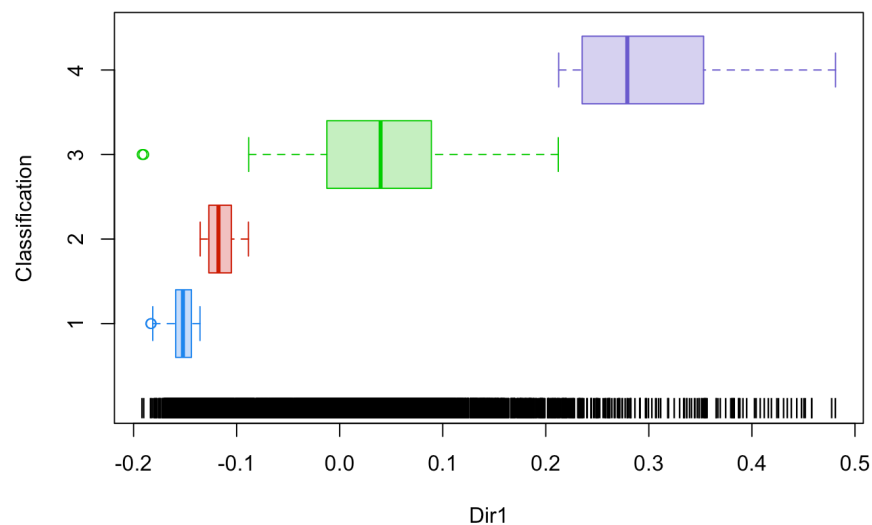


Median VAF: 0.0565
SD VAF: 0.01
VAF Range: 0.0252 - 0.073

=== Cluster 2 ===
Number of mutations: 600
Mean VAF: 0.0931
Median VAF: 0.0909
SD VAF: 0.0135
VAF Range: 0.0732 - 0.12

=== Cluster 3 ===
Number of mutations: 2823
Mean VAF: 0.2489
Median VAF: 0.2482
SD VAF: 0.0684
VAF Range: 0.0169 - 0.4206

=== Cluster 4 ===
Number of mutations: 165
Mean VAF: 0.5073
Median VAF: 0.4876
SD VAF: 0.0721
VAF Range: 0.4211 - 0.6897



The file provided, “LU-4.snv.txt”, there is no column for tumour purity, copy number, ploidy or adjusted VAF, so we cannot compute adjusted VAF we have to use raw VAF for computing CCF.

Cluster	VAF mean	Clone size (#size)	CCF	Comment
3	0.249	2823	1	Clonal (heterozygous)
4	0.507	165	1	Clonal (homozygous)
2	0.093	600	0.37	subclonal
1	0.056	619	0.23	subclonal

This implies tumour purity $\approx 2 \times 0.249 \approx 0.50$

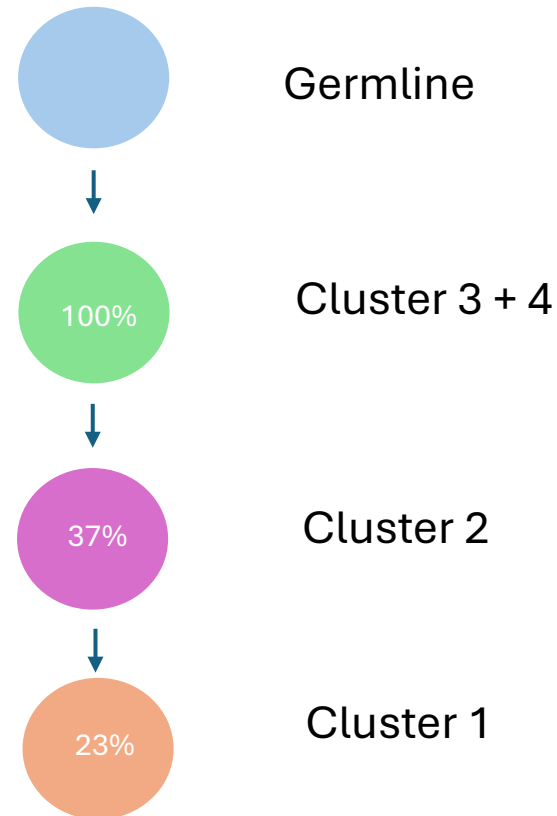
Interpretation:

From mClust, the dominant peak is Cluster 3 (mean VAF ≈ 0.249 , largest n). **Without knowing tumour purity and copy number changes, one could assume this represents a clonal heterozygous mutation in a diploid genome with uniform tumour purity, or alternatively a homozygous mutation occurring at a lower tumour purity, both of which are consistent with the observed VAF.** Under the diploid heterozygous clonal assumption, this implies tumour purity $\approx 2 \times 0.249 \approx 0.50$, and Cluster 3 is therefore classified as clonal (heterozygous) with CCF = 1.

Cluster 4 has a mean VAF ≈ 0.507 , which is approximately double the clonal heterozygous peak ($0.507 / 0.249 \approx 2$). Under this interpretation, the elevated VAF is consistent with a clonal homozygous event, loss of heterozygosity, or allele-specific amplification, and Cluster 4 is thus classified as clonal (homozygous/LOH-like) with CCF = 1.

Clusters 1 and 2 exhibit lower mean VAFs (0.056 and 0.093). When normalised to the clonal heterozygous peak, their corresponding CCFs are approximately 0.23 and 0.37, supporting their classification as subclonal populations.

3. Finally, please draw a tumour evolutionary tree to illustrate the clustering patterns listed above. Within the tree, specify clearly the cluster names/numbers and associated CCF values (**2 marks**).



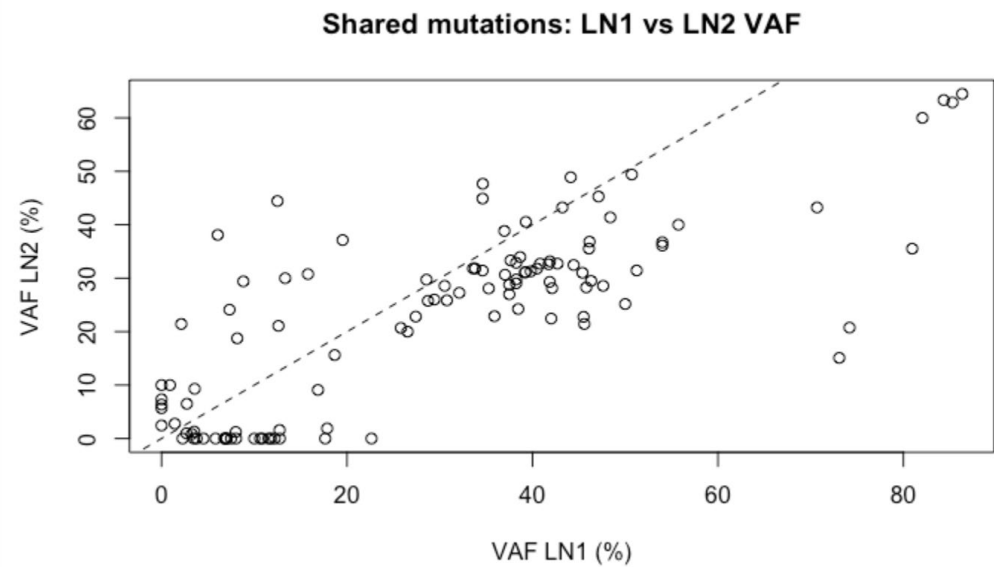
Based on Cancer Cell Fraction (CCF) estimates, Clusters 3 and 4 (CCF = 1) represent truncal mutations present in all tumour cells and form the founding/dominant clone. Cluster 2 (CCF = 0.37) represents a subclonal population that arose from the founding clone. Cluster 1 (CCF = 0.23) is a smaller subclone likely derived from Cluster 2. This hierarchical decrease in CCF supports a linear model of tumour evolution with sequential subclonal expansions.

Part 2 : Paired analysis

-
- For the paired sample analysis,
- Load the “mutations_SP4_LN1_LN2_together.txt” file, and run the mClust paired sample analysis using VAFs of all shared mutations between SP4 LN1 and LN2 samples. First, generate a scatter plot of all VAFs between LN1 and LN2 samples, and briefly comment on the plot on shared mutations between them (**2 marks**).
- Run mClust analysis based on Gaussian finite mixture model fitted by EM algorithm, followed by dimension reduction clustering and classification analysis. How many clusters can you identify? What is the size (# of mutations) for each cluster? What is the mean VAF value for each cluster in LN1 and LN2 samples, respectively? Can you identify which are major clones (heterozygous or homozygous), which are subclones, and their associated cancer cell fractions (CCF) in LN1 and LN2 samples, respectively? You can summarise all these into a table as shown in the lecture/practical slides (**4 marks**).
- Finally, please draw a tumour evolutionary tree to illustrate the clustering patterns listed above. Within the tree, specify clearly the cluster names/numbers and associated CCF values in LN1 and LN2 samples, respectively when applicable (**4 marks**).

Load the “mutations_SP4_LN1_LN2_together.txt” file, and run the mClust paired sample analysis using VAFs of all shared mutations between SP4 LN1 and LN2 samples. First, generate a scatter plot of all VAFs between LN1 and LN2 samples, and briefly comment on the plot on shared mutations between them (**2 marks**).

- The data was loaded “mutations_SP4_LN1_LN2_together.txt” file, and run the mClust paired sample analysis using VAFs of all shared mutations between SP4 LN1 and LN2 samples. (please check the LAQ4 html file)

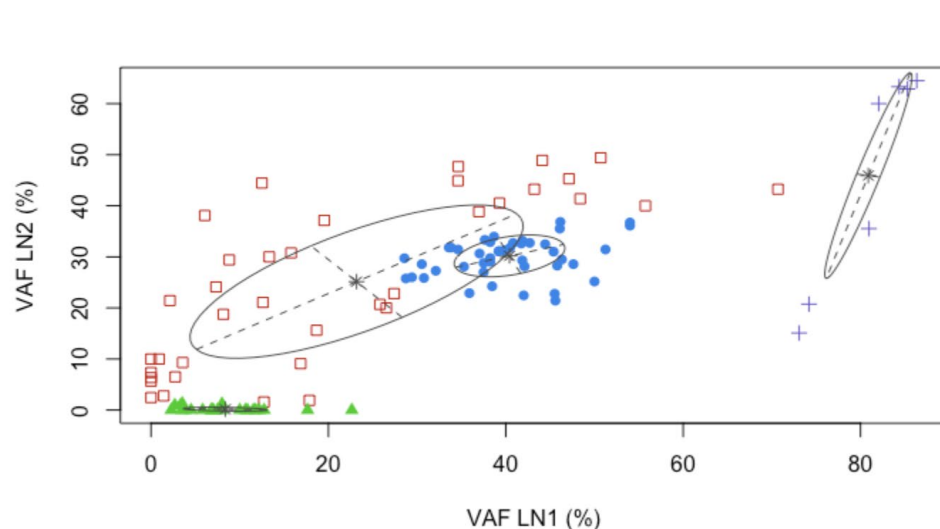


Description: df [1 x 6]					
N_shared	Mean_VAF_LN1	Mean_VAF_LN2	Median_VAF_LN1	Median_VAF_LN2	Correlation
<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
112	29.48205	22.93	32.85	27.13	0.7818143
1 row					

From the plot we can see:

1. The scatter plot shows 112 shared mutations between LN1 and LN2, with a strong positive correlation ($r = 0.78$), indicating broadly similar clonal composition across both metastatic sites. A prominent cluster of mutations at ~30–40% VAF in both samples likely represents truncal clonal events, while mutations near 0% VAF in one sample but not the other reflect subclonal divergence between the two lymph nodes. The mean VAF is higher in LN1 (29.5%) than LN2 (22.9%), and the spread around the diagonal suggests differential subclonal expansion, providing the basis for mClust clustering analysis.
2. This is exactly why we now use mClust, to formally separate these clouds. The scatter plot demonstrates a strong positive association between LN1 and LN2 VAF values ($r = 0.78$), indicating that many shared mutations have similar prevalence in both lymph node metastases. However, the noticeable spread around the diagonal suggests differential clonal expansion between the two samples.

Run mClust analysis based on Gaussian finite mixture model fitted by EM algorithm, followed by dimension reduction clustering and classification analysis. How many clusters can you identify? What is the size (# of mutations) for each cluster? What is the mean VAF value for each cluster in LN1 and LN2 samples, respectively? Can you identify which are major clones (heterozygous or homozygous), which are subclones, and their associated cancer cell fractions (CCF) in LN1 and LN2 samples, respectively? You can summarise all these into a table as shown in the lecture/practical slides (**4 marks**).



[1]	4			
1	2	3	4	
44	37	24	7	

This mClust classification plot shows shared mutations between LN1 and LN2 clustered based on their VAF values. Four distinct clusters are visible. One major cluster lies around ~40% VAF in LN1 and ~30% in LN2, representing heterozygous clonal mutations present in both metastases. A second high-frequency cluster appears at approximately ~80–85% in LN1 and ~60–65% in LN2, consistent with homozygous or LOH clonal events shared by both sites. A third, more dispersed cluster spans roughly 10–40% in LN1 and 15–45% in LN2, indicating shared subclonal populations with variable expansion between nodes. Finally, a low-frequency cluster near ~5–15% in LN1 and ~0–5% in LN2 represents minor or LN1-enriched subclonal mutations. Together, these clusters illustrate trunk mutations shared by both lymph nodes and branch-specific subclonal evolution. A total of 112 shared mutations were identified between LN1 and LN2. The mean VAF was 29.5% in LN1 and 22.9% in LN2, while the median VAFs were slightly higher at 32.9% and 27.1%, respectively, suggesting the presence of dominant clonal populations in both samples. The strong positive correlation between LN1 and LN2 VAF values ($r = 0.78$) indicates substantial similarity in clonal composition, supporting the presence of shared trunk mutations. However, the lower mean VAF in LN2 compared to LN1 suggests differences in subclonal expansion between the two metastatic sites.

Four cluster identified and characterized

##		[,1]	[,2]	[,3]	[,4]
##	[1,]	40.42042	23.17434	8.3740644	80.87664
##	[2,]	30.18589	25.14641	0.1611822	45.91522

Purity = 2 × VAF (clonal heterozygous peak)

LN1 tumour purity = 2 × 40.42% ≈ 80.8%

LN2 tumour purity = 2 × 30.19% ≈ 60.4%

Cluster	VAF LN1%	CCF LN1	VAFLN2 %	CCFLN2	Size	comment
1	40.42	1	30.19	1	44	Clonal (heterozygous)
2	23.17	0.57	25.15	0.83	37	Subclonal in both
3	8.37	0.21	0.16	0.0053	24	Subclonal in LN1, minimal in LN2
4	80.87	1 (capped value)	45.91	1 (capped value)	7	Clonal (homozygous)

Finally, please draw a tumour evolutionary tree to illustrate the clustering patterns listed above. Within the tree, specify clearly the cluster names/numbers and associated CCF values in LN1 and LN2 samples, respectively when applicable

